T-SQL Class Notes: Functions Classification (AdventureWorks2022 Database)

Overview

Transact-SQL (T-SQL) in SQL Server 2022 offers various functions categorized by their purpose and output, crucial for data manipulation, summarization, and ranking. Using the AdventureWorks2022 database, we explore these functions with practical examples.

Types of Functions

Rowset Functions

Rowset functions return an object that can substitute a table reference. Examples include `OPENDATASOURCE`, `OPENQUERY`, `OPENROWSET`, and `OPENXML`. These are useful for accessing external data or linked servers.

Example (AdventureWorks2022):

```
SELECT * FROM OPENROWSET('SQLNCLI',
'Server=AdventureWorksServer;Trusted_Connection=yes;', 'SELECT * FROM
AdventureWorks2022.HumanResources.Employee');
```

Scenario: Imagine pulling employee data from a remote HR system into AdventureWorks2022 for a unified report.

Aggregate Functions

Aggregate functions summarize large datasets. Examples include `SUM`, `MIN`, `MAX`, `AVG`, `COUNT`, and `COUNTBIG`. They are perfect for generating totals or averages across rows.

Example (AdventureWorks2022):

```
SELECT SUM(SalesOrderID) AS TotalOrders, AVG(TotalDue) AS AvgOrderValue
FROM AdventureWorks2022.Sales.SalesOrderHeader;
```

Scenario: A sales manager uses this to assess the total number of orders and average order value for the year 2025 up to June 23.

Ranking Functions

Ranking functions like `RANK`, `DENSE_RANK`, `NTILE`, and `ROW_NUMBER` simplify tasks such as assigning ranks or sequential numbers, enhancing data analysis.

Example (AdventureWorks2022):

```
SELECT BusinessEntityID, SalesYTD, ROW_NUMBER() OVER (ORDER BY SalesYTD DESC) AS
SalesRank
FROM AdventureWorks2022.Sales.SalesPerson;
```

Scenario: A company ranks salespeople based on year-to-date sales to identify top performers as of 12:32 PM WAT on June 23, 2025.

Scalar Functions

Scalar functions take a single value as input and return a single value, useful for data transformation. Examples include `CONVERT`, `GETDATE`, `ROUND`, and `SUBSTRING`.

Example (AdventureWorks2022):

```
SELECT BusinessEntityID, CONVERT(varchar, OrderDate, 107) AS FormattedDate
FROM AdventureWorks2022.Sales.SalesOrderHeader;
```

Scenario: Formatting order dates from the SalesOrderHeader table into a readable "Mon dd, yyyy" format for a customer report.

Classwork

1. **Rowset Function Task**: Use `OPENQUERY` to fetch employee details from a linked server 'AdventureWorksLinked' with the query 'SELECT * FROM HumanResources.Employee'.
2. **Aggregate Function Task**: Calculate the total and average sales amount from `Sales.SalesOrderHeader` using `SUM` and `AVG`.
3. **Ranking Function Task**: Assign a rank to products in `Production.Product` based on `ListPrice` using `DENSE_RANK()`.
4. **Scalar Function Task**: Convert the `OrderDate` in `Sales.SalesOrderHeader` to a string in 'yyyy-mm-dd' format using `CONVERT`.

Tests

Test 1: Rowset Function

Write a query using `OPENROWSET` to retrieve the top 5 employees by hire date from `HumanResources.Employee`.

Expected Output Example:

```
BusinessEntityID: 1, HireDate: 2009-01-14
```

Test 2: Aggregate Function Combination

Write a query to find the total, average, and maximum `TotalDue` from `Sales.SalesOrderHeader` for orders placed in 2025.

Expected Output Example:

```
TotalDue: 150000, AvgDue: 750, MaxDue: 5000
```

Test 3: Ranking Function

Use `NTILE(4)` to divide salespeople in `Sales.SalesPerson` into 4 quartiles based on `SalesYTD`.

Expected Output Example:

```
BusinessEntityID: 274, SalesYTD: 5596976.14, Quartile: 1
```

Test 4: Scalar Function

Extract the year from `OrderDate` in `Sales.SalesOrderHeader` using `YEAR` and concatenate it with ' Orders' using `+`.

Expected Output Example:

```
OrderYear: 2025 Orders
```

Additional Notes

- Rowset functions are ideal for integrating external data, like merging AdventureWorks2022 with a supplier database.
- Aggregate functions help in financial reporting, e.g., quarterly sales summaries.
- Ranking functions assist in performance evaluations or inventory prioritization.
- Scalar functions enhance data presentation, such as formatting dates for invoices.

Cheatsheet

Function Type	Examples	Explanation
Rowset Functions	`OPENDATASOURCE`, `OPENQUERY`, `OPENROWSET`, `OPENXML`	Return a table-like object for external data access, e.g., linking to HR systems.
Aggregate Functions	`SUM`, `MIN`, `MAX`, `AVG`, `COUNT`, `COUNTBIG`	Summarize data across rows, e.g., total sales or average order value in SalesOrderHeader.
Ranking Functions	`RANK`, `DENSE_RANK`, `NTILE`, `ROW_NUMBER`	Assign ranks or sequential numbers, e.g., ranking salespeople by SalesYTD.

Function Type	Examples	Explanation
Scalar Functions	`CONVERT`, `GETDATE`, `ROUND`, `SUBSTRING`	Transform single values, e.g., formatting OrderDate or extracting substrings.